Computational Physics Simulations with ALPS

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Introduction and Motivation

- ALPS, Algorithms and Libraries for Physics Simulations, is an open-source software package used to run computational physics simulations
- Tutorials cover Monte Carlo(MC) simulations, Exact Diagonalization(ED), Density Matrix Renormalization Group(DMRG), and related numerical approaches
- Makes simulation of complex quantum systems accessible and reproducible

ALPS Software Package

Personal Contributions

- Supported ALPS deployment across heterogeneous systems via binary and Spack installs, including the Virginia Tech Falcon Cluster.
- Produced instructional videos for Spack and Falcon Cluster installation
- Executed and analyzed ALPS physics simulations

Lattice Systems

- 1-Dim Chain

Figure 1. One-dimensional chain lattice of L sites $\vec{S}_i$ connected by nearest-neighbor bonds, with periodic boundary conditions

- Two-Leg Ladder

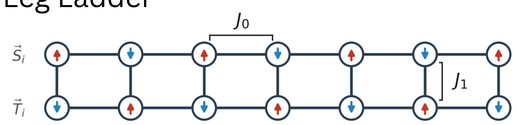

Figure 2. Two-leg ladder lattice formed by two parallel chains $\vec{S}_i$ and $\vec{T}_i$ of L sites each, connected by rung bonds, with periodic boundary conditions

Magnetization in Heisenberg Chain and Ladder Systems

Goal of Simulation: Compare magnetization curves m(h) of the Heisenberg chain and ladder to characterize the effect of rung coupling

Observables: $\langle m \rangle = \frac{1}{L}\sum_i \langle S_i^z \rangle, \qquad \langle S_i^z \rangle = \frac{1}{Z}\,\mathrm{Tr}\left(e^{-H/T}\,S_i^z\right)$

Heisenberg Chain (S=1/2)

Parameters:
- System size: L=20
- Coupling: J=1
- Temperature: T/J=0.08
- Field range: h/J∈[0,2.5]
- Thermalization: 1,000 sweeps
- Measurement: 5,000 sweeps

Background :
- 1-Dim array of spin-½ particles with nearest-neighbor exchange interactions
- Magnetic excitations on the chain are called spinons (spin-½ quasiparticles)
- Simplest non-trivial quantum many-body system
- Models real 1D magnetic materials like Sr_2CuO_3 and $KCuF_3$
- Has an exact analytical solution, Bethe ansatz

Hamiltonian:

$$H = \sum_{i=1}^{L} \vec{S}_i \cdot \vec{S}_{i+1} + h \sum_{i=1}^{L} S_i^z$$

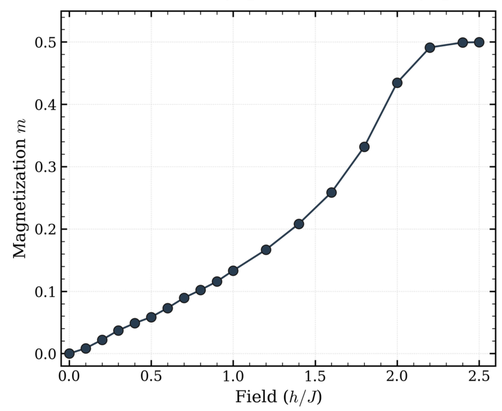

Figure 3. Magnetization m vs. field h/J for the spin-½ Heisenberg chain.

Heisenberg Ladder (S=1/2)

Hamiltonian:

$$H = \sum_i \left[J_0 \left(\vec{S}_i \cdot \vec{S}_{i+1} + \vec{T}_i \cdot \vec{T}_{i+1} \right) + J_1\, \vec{S}_i \cdot \vec{T}_i + h\left(S_i^z + T_i^z\right) \right]$$

Parameters:
- System size: L=20
- Width: W=2
- Coupling: $J_0 = J_1 = 1$
- Temperature: T/J=0.08
- Field range: h/J∈[0,2.5]
- Thermalization: 1,000 sweeps
- Measurement: 5,000 sweeps

Background :
- Two coupled spin- ½ chains with leg coupling J_0 and rung coupling J_1
- Magnetic excitations are bound spinon pairs created by rung coupling
- Quasi-1D quantum many-body system with no exact analytical solution

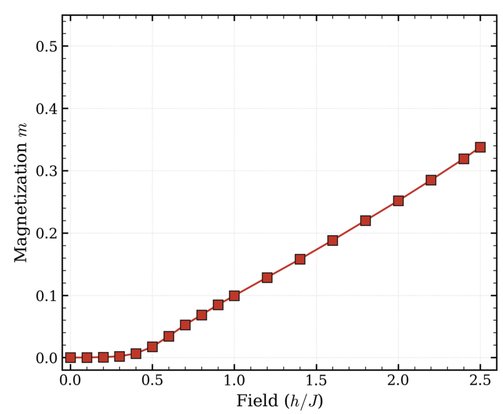

Figure 4. Magnetization m vs. field h/J for the spin-½ two-leg Heisenberg ladder.

Comparison and Discussion

- Both systems are antiferromagnetic spin-½ networks simulated using path-integral Quantum Monte Carlo
- Unlike the chain, the ladder has no exact solution, making numerical simulation essential
- Chain spectrum is gapless, allowing arbitrarily low-energy excitations, while the ladder's rung coupling opens a finite spin gap that the field must overcome
- The overlay isolates the rung coupling's effect: this single structural difference produces the entire low-field gap

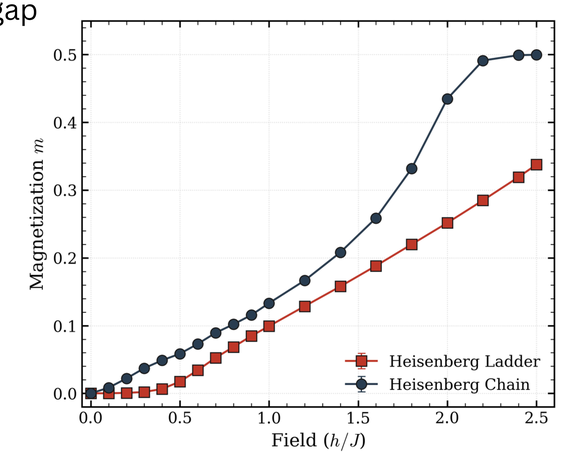

Figure 5. Comparison of Magnetization m vs. field h/J for the spin-½ Heisenberg chain and ladder

Future Directions

- Apply QMC to system sizes and temperatures inaccessible to ED and DMRG
- Expand instructional and deployment resources
- Continued NSF support (Award No. 2345575)

Acknowledgements

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- Full references and resources available via QR code